Seongyoon Kim

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EDUCATION

University of Illinois Urbana-Champaign Expected 2027
Doctor of Philosophy in Mathematics

University of Illinois Urbana-Champaign 2022
Master of Science in Mathematics

Sungkyunkwan University 2019
Bachelor of Science in Economics

Erasmus University Rotterdam 2018
Bachelor of Science in Economics and Business Economics

QUANTITATIVE SKILLS

Quantitative Methods: Stochastic Processes, Stochastic Differential Equations, Time-Series Analysis, Optimization, Financial Econometrics, Mathematical Finance, Risk Modeling & Management
Programming: Python: NumPy, pandas, scikit-learn, PyTorch, SQL

SELECTED QUANTITATIVE RESEARCH

Predicting Price Surges with Neural ODE 2026
Manuscript under review

- Developed a Neural ODE-based classifier for predicting rare price surges in low signal-to-noise financial time series
- Derived a closed-form first-passage-time distribution for a generalized jump-diffusion process to construct a stochastic surge-classification benchmark.
- Evaluated the models on Korean and U.S. equity data against LSTM, CNN, Transformer, and jump-diffusion benchmarks, with the Neural ODE achieving the highest average precision across the tested datasets.

On the Expansion of Risk Pooling 2026
Management Science

- Modeled participant losses as random variables in an expanding risk pool, applied entropic risk measures and equilibrium-pricing theory to derive participation and stability conditions
- Implemented Python-based numerical experiments with real insurance data to evaluate expansion and reinsurance scenarios.

Tokenization of Distributed Insurance by Auction 2024
Japanese Journal of Statistics and Data Science

- Designed an auction-based allocation framework using constrained knapsack optimization and analyzed how strategic bid merging and splitting affect investor allocation, returns, and market efficiency.

SELECTED EXPERIENCE

Graduate Research & Teaching Assistant 2020–2024; 2026–Present
University of Illinois Urbana-Champaign, Department of Mathematics

- Developed theoretical and quantitative models in risk sharing, prediction, pricing, and uncertainty management.
- Served as a teaching assistant and discussion instructor for courses in probability, stochastic processes, machine learning, calculus, and linear algebra.

Mandatory Military Service, Driver 2025–2026
Republic of Korea Army, Mungyeong, South Korea

HONORS & PRESENTATIONS

PARM Research Fellow, University of Illinois Urbana-Champaign 2024
Presenter, International Conference on Insurance & Actuarial Science, Nankai University, Tianjin 2024
Presenter, 2022 INFORMS Annual Meeting, Indianapolis, IN 2022